

Senior Embedded Linux & Kernel Engineer

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[Blog](#) | [Github](#) | [Gitlab](#) | [GnuPG](#) | [Linkedin](#)

Embedded Linux and low-level software engineer with 15 years of experience bringing up platforms and delivering production systems. Expertise across Linux kernel and BSP development, boot chains, Yocto/OpenEmbedded, secure systems, and over-the-air updates for ARM, PowerPC, and RISC-V platforms. Open-source contributor to Linux, U-Boot, OpenEmbedded/Yocto, and Buildroot.

Technical Expertise

Linux and build systems: Linux kernel, Yocto, OpenEmbedded, Buildroot, RIOT-OS, Zephyr

Boot and security: U-Boot, TF-A, AT91, OP-TEE, TPM, trusted and measured boot, encryption, integrity systems

Platforms: ARMv7, ARMv8, PowerPC, RISC-V, x86-64; QorIQ, ARMADA, i.MX, STM32MP1, SAMA5, SAMA7, Tegra, U740, X280

Development: C, assembly, Python, Bash, Git, repo, kas, Gerrit

Debugging and tools: QEMU, GDB, Valgrind, Lauterbach, Saleae, Podman

Standards: DO-178B DAL A, EABI, ISO C99, POSIX

Languages

French: Native

English: Professional working proficiency

Education

2011: Master's degree (with honors), Architecture of critical and real-time embedded systems

Additional Information

Linux technologist and free-software enthusiast. Personal interests include robotics, IoT, retro computing, arcade systems, and CrossFit.

Experiences

December 2020 - Present: Embedded Linux & Kernel Engineer

Bootlin, Toulouse

- Port Linux to new platforms and develop device drivers.
- Integrate open-source components and build complete embedded Linux systems.
- Contribute board support, device drivers, and build-system changes upstream to Linux, U-Boot, OpenEmbedded/Yocto, and Buildroot.

July 2015 - December 2020: Embedded Linux Engineer

SIGFOX

- Performed board bring-up and developed BSPs, including bootloaders and Linux kernels.
- Delivered boot flows, partitioning, OTA update, and rollback mechanisms.
- Developed security components including TPM integration, encryption, integrity systems, trusted boot, and measured boot.

April 2014 - July 2015: Android System Integrator

Intel Android SI NPI, via Celad & SHT, Toulouse

- Introduced and maintained pre-silicon platforms in Android mainline.
- Published releases and documented known-good configurations and methods.
- Provided bring-up, build, first-boot, and power-on support.

February 2014 - March 2014: ROS System Integrator

Sogeti High Tech, R&D, Toulouse

- Designed an embedded ROS node with ROSc for coOS.
- Designed a custom ROS bridge for the Unicom framework.

August 2011 - February 2014: Embedded system software engineer

Airbus S.A.S, via Sogeti High Tech, Toulouse

- Developed and maintained primary flight-control software for Airbus A380 and A400M aircraft.
- Optimized embedded-code WCET and performed certification analysis.

February 2011 - August 2011: Research Assistant / ROS System Developer

Five-month contract at **Office National d'Études et de Recherches Aéronautiques**, Toulouse

- Designed and implemented a ROS tool to simulate communication between different software modules of heterogeneous critical systems.

October 2010 - February 2011: Internship / VHDL Processor Designer

Five-month internship at **Atelier Interuniversitaire de Micro-nano Électronique**, Toulouse

- Designed a 32-bit scalar, pipelined, in-order RISC processor with a Harvard architecture.

February 2010 - August 2010: Research Assistant - WCET Analysis

Six-month contract at **Institut de Recherche en Informatique de Toulouse**, Toulouse

- Performed timing analysis of parallel processes on multi-core processors.